Answer Key & Teacher Notes

Session 1 — Graphing Inequalities · Session 2 — Testing Solutions to Inequalities

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Graphing Inequalities

TALK PROMPTS

- How does the dot at the boundary change what the graph is telling you?
- How would the graph change if the rule said "below 20 degrees" instead of "20 degrees or lower"?

PARTNER TASK

Roller Coaster Height. One partner writes $h \geq 52$ and graphs it; the other tests 51, 52 and 60 and says which are under the arrow. Switch roles with the freezer rule $t \leq 8$.

WATCH FOR

- Choosing the dot from the sign of the number (negative means open) instead of from the symbol.
- Pointing the arrow the wrong way for ≥ or ≤ (the arrow points toward the numbers that make the statement true).
- Using a closed dot for "under" or "more than," which leave the boundary out.

SESSION 2 · TEACHER NOTES

Testing Solutions to Inequalities

TALK PROMPTS

- Why is the boundary number the most useful value to test?
- Why can a value be very close to the boundary and still not be a solution?

PARTNER TASK

Zip Line. A zip line lets a rider go only when the rider weighs more than 70 pounds. One partner writes $w > 70$ and tests 70 and 85; the other graphs it and explains the dot. Switch with the Backpack Rule: "at most 30 pounds."

WATCH FOR

- Thinking the boundary is always a solution (it is only for ≤ and ≥).
- Reading "more than 45" as $n \geq 45$.
- Listing only whole numbers as solutions; decimals and negative numbers can work too.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.B.8	1	2	1	1	1	2	2	3	2	1	1
S1 · Form B	6.EE.B.8	1	2	1	1	1	2	2	3	2	1	1
S2 · Form A	6.EE.B.8	1	2	1	1	2	2	2	3	2	1	1
S2 · Form B	6.EE.B.8	1	2	1	1	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

boundary point: The number where the solutions **start** or stop.

closed circle: A filled dot: the boundary **is** a solution.

open circle: An empty dot: the boundary is **not** a solution.

solution set: **All** the numbers that make an inequality true.

- Use a **closed** dot for ≤ and ≥. Use an **open** dot for < and >.
- Draw the arrow toward the numbers that make the inequality **true**.

Example: The boundary number is **6**. < does not include 6, so draw an **open** dot at 6. *x* is less than 6, so the arrow points **left**. Answer: Check: 5 < 6 is **true**, and 5 is under the arrow.

YOU TRY

1. closed dot at −2, arrow right

2. a. $x \leq 4$ b. $x > -1$

EXIT TICKET

1. a. open dot at 3, arrow right b. closed dot at −2, arrow left

2. $d \leq 40$; closed dot at 40, arrow left

3. **C** ($x < 5$)

PRACTICE FORM A

1. a. open, right b. closed, left c. closed, right d. open, left

2. a. $w \geq 15$ b. closed dot at 15, arrow right c. Sample: 20 lb works; 12 lb does not

3. $t \leq 8$; closed dot at 8, arrow left (toward colder temperatures)

4. $t \leq 20$; closed dot at 20, arrow left

5. $b \geq 18$; closed dot at 18, arrow right

6. a. $x \geq -3$ b. Yes. The closed dot shows −3 is included.

7. a. $p \geq 250$; closed dot at 250, arrow right b. Sample totals: 250 and 300 (any total 250 or more)

8. a. $d \geq 4$ b. $w < 200$ c. "At least 4" includes 4, so its dot is closed. "Under 200" leaves out 200, so its dot is open.

9. The closed dot is right, but ≥ means 30 or greater, so the arrow must point right. Correct: closed dot at 30, arrow right. Check: 35 ≥ 30 is true.

10. **B** (a closed dot at 5, arrow right)

11. **B** (not a solution)

Self-check word **ARROWS**

PRACTICE FORM B

1. a. closed, left b. open, right c. open, left d. closed, right

2. a. $h \geq 52$ b. closed dot at 52, arrow right c. Sample: 55 in. works; 50 in. does not

3. $b \leq 36$; closed dot at 36, arrow left (toward fewer bikes)

4. $t \leq 65$; closed dot at 65, arrow left

5. $a \geq 12$; closed dot at 12, arrow right

6. a. $x < 2$ b. No. The open dot shows 2 is not included.

7. a. $c \geq 180$; closed dot at 180, arrow right b. Sample totals: 180 and 225 (any total 180 or more)

8. a. $h \geq 48$ b. $w < 10$ c. "At least 48" includes 48, so its dot is closed. "Under 10" leaves out 10, so its dot is open.

9. The closed dot is right, but ≤ means 7 or less, so the arrow must point left. Correct: closed dot at 7, arrow left. Check: 4 ≤ 7 is true.

10. **C** (an open dot at 8, arrow left)

11. **A** (a solution)

Self-check word **CIRCLE**

GUIDED NOTES

substitute: Put a number in **place** of the variable.

solution: A value that makes the inequality **true**.

true statement: A comparison that is **correct**.

- To test a value, substitute it and ask whether the statement is **true**.
- For ≤ and ≥, the boundary **is** a solution. For < and >, it is **not**.

Example: 8 < 9 is **true**, so 8 is a solution. 9 < 9 is **false**, so the dot at 9 is **open**. 10 < 9 is false, so 10 is not a solution. Answer: Only **8** works. Graph: open dot at 9, arrow left.

YOU TRY

1. a. yes b. no

2. −1 > 2 false; 2 > 2 false; 5 > 2 true. Only 5. Graph: open dot at 2, arrow right.

EXIT TICKET

1. a. no b. yes

2. v < 85; 85 < 85 is false (not a solution); 60 < 85 is true (solution)

3. **B** (−3)

PRACTICE FORM A

1. a. yes b. no c. no d. no e. yes f. yes

2. a. w > 80 b. 95 > 80 is true, so 95 is a solution c. 80 > 80 is false, so 80 is not a solution d. open dot at 80, arrow right

3. n > 45; 45 > 45 is false (not a solution); 52 > 45 is true (solution)

4. c < 25; 25 does not; 19 works

5. a > 15; no, 15 > 15 is false

6. 5 ≤ 5 is true (solution); 6 ≤ 5 is false (not a solution). Graph: closed dot at 5, arrow left.

7. w < 12; 12 does not; 9 works; 15 does not

8. a. w ≤ 50 b. 50 and 38 are allowed; 51 is not c. closed dot at 50, arrow left; the closed dot shows a bag of exactly 50 pounds is allowed

9. 26 > 26 is false, so 26 is not a solution. The dot must be open: open dot at 26, arrow right.

10. **D** (8.5)

11. **B** (whether the dot is open or closed)

Self-check word **VALUES**

PRACTICE FORM B

1. a. yes b. no c. yes d. no e. yes f. no

2. a. w > 70 b. 85 > 70 is true, so 85 is a solution c. 70 > 70 is false, so 70 is not a solution d. open dot at 70, arrow right

3. n > 30; 30 > 30 is false (not a solution); 41 > 30 is true (solution)

4. c < 28; 28 does not; 24 works

5. a > 12; no, 12 > 12 is false

6. 4 ≥ 4 is true (solution); 3 ≥ 4 is false (not a solution). Graph: closed dot at 4, arrow right.

7. d < 400; 400 does not; 350 works; 410 does not

8. a. w ≤ 30 b. 30 and 24 are allowed; 32 is not c. closed dot at 30, arrow left; the closed dot shows a pack of exactly 30 pounds is allowed

9. 15 < 15 is false, so 15 is not a solution. The dot must be open: open dot at 15, arrow left.

10. **C** (7)

11. **A** (the boundary value)

Self-check word **CHECKS**

SHEET 1 · GRAPHING INEQUALITIES

1. a. closed, right b. open, left c. closed, left d. open, right e. closed, right f. open, left g. open, right h. closed, left
2. Closed dot at −1, arrow left. Sample solution: −3 (any number −1 or less).
3. $a \geq 10$. Closed dot at 10, arrow right.
4. $m < 6$. Sample: A small boat can hold fewer than 6 people.

SHEET 2 · TESTING SOLUTIONS TO INEQUALITIES

1. $x > 7$: no, no, yes. $x \leq 7$: yes, yes, no. $x \geq 4$: yes, yes, yes.
2. a. no ($9 < 9$ is false) b. yes ($0 \geq -1$) c. no ($3 \leq 2.5$ is false)
3. $s > 450$. $450 > 450$ is false; $445 > 450$ is false; $472 > 450$ is true. Only 472 wins.
4. $-3 \leq -1$ is true; $0 \leq -1$ is false; $2 \leq -1$ is false. Only −3 is a solution. Graph: closed dot at −1, arrow left.
5. Every number less than 12 is a solution, including decimals like 11.5 and negative numbers like −3. There are infinitely many solutions.
6. **A** (−3)
7. **D** (15.5)

October 2026 conceptual additions — teacher key

Session 1 · Graphing Inequalities

Closed circle at −2, ray right. −3 fails; −2 and 0 satisfy. Closed circle includes the boundary.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Closed circle at −2, ray right. −3 fails; −2 and 0 satisfy. Closed circle includes the boundary.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Testing Solutions to Inequalities

−1 satisfies; 1.5 and 2 fail. Example 1.25 also satisfies. Every real number less than 1.5 belongs, infinitely many values.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: −1 satisfies; 1.5 and 2 fail. Example 1.25 also satisfies. Every real number less than 1.5 belongs, infinitely many values.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.